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Chaotic Dynamics of Tai Chi Public Attention Revealed by an Integrated Framework of Horizontal Visibility Graphs, Autoencoders, and Sparse Identification

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ABSTRACT

Understanding the dynamical patterns of public attention towards traditional cultural practices such as Tai Chi is crucial for cultural heritage preservation and promotion strategies. This study reveals the chaotic dynamics underlying Tai Chi public attention through an integrated analytical framework combining Horizontal Visibility Graphs (HVG), Autoencoder neural networks, and Sparse Identification of Nonlinear Dynamics (SINDy). Using daily Baidu Index data (2014–2024) from four representative Chinese provinces (Beijing, Shanghai, Guangdong, Henan), we construct HVGs and quantify chaos through degree distribution exponents (λ). Results reveal significant regional heterogeneity: HVG λ ranges from 0.291 (Beijing PC, strong chaos) to 0.437 (Shanghai PC, quasi-periodic), with corresponding maximum Lyapunov exponents of 0.025–0.036 bits/day (Rosenstein method) and 0.016–0.217 bits/day (Wolf method); the consistent positivity across both estimation methods provides cross-validated confirmation of chaotic dynamics. Autoencoder-based dimensionality reduction achieves reconstruction correlations of 0.896–0.918, enabling discrete SINDy to identify sparse governing equations (37–38 active terms from 48 candidates) with normalized root-mean-square errors of 11.7–12.8%. The dynamical conclusions are triangulated across three analytically independent characterizations — HVG topological classification, Lyapunov-based phase space analysis, and SINDy equation structure — which converge on consistent regional rankings, with Beijing exhibiting the strongest chaotic signatures and Henan the most structured quasi-periodic behavior. Phase space analysis confirms diverse attractor geometries: Beijing exhibits space-filling chaotic trajectories while Shanghai displays quasi-periodic structures. Comparative evaluation against ARIMA, VAR, and LSTM baselines within the same latent space confirms that SINDy achieves superior predictive fidelity (NRMSE 17.6–22.4%) while uniquely providing interpretable governing equations and chaos diagnostics inaccessible to alternative approaches. This interdisciplinary framework bridges complex network theory, deep learning, and dynamical systems analysis, offering a rigorous quantitative paradigm for studying the temporal evolution of public attention in cultural phenomena.

Introduction

Understanding the dynamical patterns hidden in complex time series is essential for explaining how social and cultural events evolve over time. Compared with many physical signals, attention-related series are often shaped by intertwined mechanisms—long-term trend shifts, seasonal cycles, media amplification, and collective responses—so their evolution is typically nonlinear and may even exhibit chaotic characteristics. This complexity makes it difficult for traditional linear tools to capture regime changes or to describe the intrinsic rules that generate observed fluctuations^{1–3}, a challenge that has been widely recognized in the study of social dynamics more broadly⁴. Therefore, methods that can both characterize nonlinear structure and infer plausible dynamical mechanisms from data are increasingly needed.

Against this background, Tai Chi offers a meaningful cultural object whose public attention dynamics can be quantified. Originating in the mid-17th century in Henan Province of China, Tai Chi has gradually become a widely practiced cultural activity. In December 2020, Tai Chi was included in the UNESCO list of representative works of human intangible cultural heritage, highlighting its universal cultural value and attracting more than 100 million practitioners from more than 150 countries around the world^{5,6}. Nevertheless, public attention to Tai Chi is not driven by a single event; it reflects the combined influence of policy initiatives, media coverage, seasonal rhythms, and regional cultural foundations. The resulting daily search-index series thus provides an appropriate empirical basis for investigating how cultural attention evolves under multi-factor interactions and

heterogeneous local contexts⁷.

However, understanding the time-varying dynamics of such cultural phenomena requires the development of appropriate analytical frameworks. While traditional methods like linear regression or Fourier analysis fail to capture the complex and nonlinear nature of public engagement, recent advancements in dynamical systems and data-driven techniques have shown promise in addressing these challenges. For instance, the Horizontal Visibility Graph (HVG) method offers a way to map time series into network structures, enabling an exploration of temporal dependencies and deviations from randomness^{8–10}. More broadly, complex network representations of multivariate time series have demonstrated capacity to reveal dynamical structure across diverse physical and social systems^{3,11}. However, HVG-based methods primarily focus on characterizing the structure of the time series rather than providing explicit models of the underlying dynamics^{12,13}. Similarly, Sparse Identification of Nonlinear Dynamics (SINDy) has proven effective in discovering governing equations from time series^{14–16}, but it is limited when dealing with high-dimensional data or when the appropriate state variables are not immediately apparent—a challenge that has motivated recent efforts to combine SINDy with deep learning architectures for automated coordinate discovery^{17,18}. Despite these advances, no existing framework integrates these methods to both characterize and model the complex dynamics underlying phenomena like Tai Chi's public attention^{19,20}. The resilience and adaptability of such attention dynamics, moreover, remain poorly understood from a dynamical systems perspective²¹.

To address these limitations, this study proposes an integrated approach that combines the strengths of HVG, SINDy, and autoencoders. By leveraging HVG-based network features, we can capture the temporal structure of Tai Chi's public attention. Using the SINDy framework, we can identify sparse governing equations that describe the evolution of attention over time, while autoencoders allow us to reduce the dimensionality of the data and uncover latent variables that better represent the underlying dynamics. Our approach thus aims to provide a comprehensive understanding of Tai Chi's public attention dynamics by combining network analysis, dynamical modeling, and data reduction. Specifically, we apply this integrated method to daily search index data from four Chinese provinces (Beijing, Shanghai, Guangdong, and Henan) over a ten-year period, enabling us to investigate regional variations in the temporal patterns of Tai Chi engagement.

Methods and Datasets

This study proposes an integrated HVG–Autoencoder–SINDy framework built around a sequential analytical logic. The HVG transforms each raw time series into a network representation whose degree distribution encodes the temporal correlation structure of the underlying dynamics, and extracts a 12-dimensional topological feature vector per sliding window that captures local dynamical properties across time. Because these feature vectors are high-dimensional and the relevant state variables of the underlying system are not directly observable, an autoencoder is employed to compress them into a low-dimensional latent space, where the learned representation retains the essential dynamical structure while remaining tractable for subsequent equation discovery. SINDy then operates on the latent trajectories to identify sparse governing equations, exploiting the compact and geometrically meaningful coordinates provided by the autoencoder to make reliable sparse regression feasible. Classical chaos analysis tools — Lyapunov exponent estimation, phase space reconstruction, and Poincaré sections — are employed alongside this pipeline to independently validate the dynamical characterization obtained from the HVG analysis. The overall architecture of this pipeline is illustrated in Figure 0.

Horizontal Visibility Graph

The HVG algorithm maps a time series $\{x_t\}_{t=1}^N$ to an undirected graph $G = (V, E)$ where each time point corresponds to a node⁸. Two nodes i and j (with $i < j$) are connected by an edge if and only if all intermediate values satisfy the horizontal visibility criterion:

$$x_k < \min(x_i, x_j), \quad \forall k : i < k < j$$

This geometric condition ensures that a horizontal line connecting (i, x_i) and (j, x_j) lies entirely above all intermediate points. The resulting graph inherits dynamical properties of the original time series through its topological structure⁸.

A fundamental theoretical result establishes that for uncorrelated random time series, the HVG degree distribution follows an exponential decay $P(k) = \frac{1}{3} \left(\frac{2}{3}\right)^{k-2}$, equivalently expressed as:

$$P(k) \sim \exp(-\lambda k), \quad \lambda_{\text{random}} = \ln(3/2) \approx 0.405$$

This universal exponent serves as a reference for chaos detection. Luque et al.⁸ proved analytically that for i.i.d. random series $P(k) = \frac{1}{3} \left(\frac{2}{3}\right)^{k-2}$, yielding $\lambda_{\text{random}} = \ln(3/2) \approx 0.4055$. The mechanistic interpretation is transparent: chaotic time series possess long-range temporal correlations that elevate the probability of distant time points satisfying the horizontal visibility criterion, thereby increasing the frequency of large-degree nodes and flattening the degree distribution below the random baseline ($\lambda < \ln(3/2)$); conversely, periodic series concentrate visibility connections locally, steepening the distribution

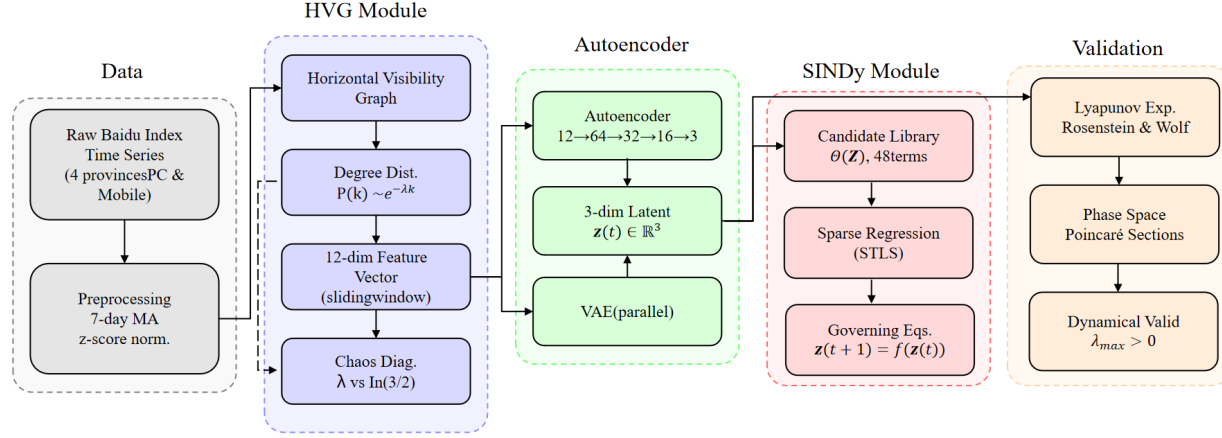


Figure 0. Overall framework of the integrated HVG–Autoencoder–SINDy analytical pipeline. Preprocessed time series are passed sequentially through the HVG module (feature extraction), autoencoder (dimensionality reduction to a 3-dimensional latent space), and SINDy (sparse governing equation discovery). Classical chaos diagnostics are applied in parallel for independent validation.

($\lambda > \ln(3/2)$). This boundary has been analytically validated for the logistic map, Lorenz attractor, and fractional Brownian motion, and is now established as a standard criterion for HVG-based dynamical classification^{22–25}.

The parameter λ is estimated by applying OLS linear regression in the log-transformed domain: for each empirical degree probability $P(k) > 0$, we fit $\log P(k) = a - \lambda k$ via `numpy.polyfit` (Python package), recovering λ as the negative slope. This semi-logarithmic regression is the standard approach for exponential distribution fitting, as it renders the model linear in its parameters and ensures unbiased slope estimation under the exponential decay assumption.

To capture the temporal evolution of network properties, we employ a sliding window approach. For each window of length W , we construct the HVG and extract a 12-dimensional feature vector comprising: degree statistics (mean, standard deviation, maximum, minimum), local clustering coefficient (mean, standard deviation), network density, local λ estimate, and time series statistics within the window (mean, standard deviation, skewness, kurtosis). These feature vectors form the input to the subsequent autoencoder.

Autoencoder Architecture

An autoencoder learns a compressed latent representation through an encoder-decoder structure trained to minimize reconstruction error²⁶. Given input feature vectors $\mathbf{x} \in \mathbb{R}^d$, the encoder $f_\phi : \mathbb{R}^d \rightarrow \mathbb{R}^m$ maps to a lower-dimensional latent space ($m \ll d$), and the decoder $g_\theta : \mathbb{R}^m \rightarrow \mathbb{R}^d$ reconstructs the original input:

$$\mathbf{z} = f_\phi(\mathbf{x}), \quad \hat{\mathbf{x}} = g_\theta(\mathbf{z})$$

The network parameters ϕ, θ are optimized by minimizing the mean squared reconstruction error:

$$\mathcal{L}_{\text{AE}} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2$$

We implement a fully connected architecture with encoder layers $[12 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 3]$ and symmetric decoder layers $[3 \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow 12]$, using ReLU activations and batch normalization. The 3-dimensional latent space facilitates visualization and subsequent dynamical analysis.

We additionally consider a Variational Autoencoder (VAE) variant that imposes a probabilistic structure on the latent space²⁷. The encoder outputs parameters of a Gaussian distribution $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mu, \text{diag}(\sigma^2))$, and latent samples are drawn via the reparameterization trick: $\mathbf{z} = \mu + \sigma \odot \epsilon$, where $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The VAE loss combines reconstruction error with a KL-divergence regularizer:

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{recon}} + \beta \cdot D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))$$

where $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ is the prior and β controls the regularization strength.

Sparse Identification of Nonlinear Dynamics

Given latent trajectories $\mathbf{z}(t) \in \mathbb{R}^m$ from the autoencoder, SINDy discovers sparse governing equations by regressing derivatives against a library of candidate functions²⁸. For discrete-time dynamics appropriate to our daily time series, we seek equations of the form:

$$\mathbf{z}(t+1) = \mathbf{f}(\mathbf{z}(t))$$

We construct a library matrix $\Theta(\mathbf{Z})$ containing candidate nonlinear functions evaluated at each time point. The library includes: constant term (1), linear terms (z_1, z_2, z_3), quadratic terms ($z_1^2, z_2^2, z_3^2, z_1 z_2, z_1 z_3, z_2 z_3$), and trigonometric terms ($\sin(z_i), \cos(z_i)$). The dynamics are then expressed as:

$$\mathbf{Z}' = \Theta(\mathbf{Z})\Xi$$

where $\mathbf{Z}'$ denotes the forward-shifted latent states and Ξ is the sparse coefficient matrix to be identified.

We employ Sequential Thresholded Least Squares (STLS) for sparse regression¹⁴. Starting from an ordinary least squares solution, coefficients below a threshold α are iteratively set to zero, and least squares is re-solved on the remaining terms until convergence¹⁶:

$$\Xi^{(k+1)} = \arg \min_{\Xi} \|\mathbf{Z}' - \Theta\Xi\|_F^2 \quad \text{s.t.} \quad \Xi_{ij} = 0 \text{ if } |\Xi_{ij}^{(k)}| < \alpha$$

The resulting sparse coefficient matrix reveals the dominant dynamical terms governing latent space evolution, yielding interpretable equations with typically 3–6 active terms per dimension.

Lyapunov Exponent Estimation

The largest Lyapunov exponent $\lambda_{\max}$ quantifies the exponential rate of divergence between nearby trajectories, serving as the definitive chaos indicator: $\lambda_{\max} > 0$ implies sensitive dependence on initial conditions (chaos), while $\lambda_{\max} \leq 0$ indicates regular dynamics²⁹.

We employ the Rosenstein algorithm³⁰, which tracks the average logarithmic divergence of nearest neighbors in reconstructed phase space. Given a time series x_t , the phase space is reconstructed via delay embedding:

$$\mathbf{y}(t) = [x(t), x(t+\tau), x(t+2\tau), \dots, x(t+(m-1)\tau)]$$

where m is the embedding dimension and τ is the time delay. For each point $\mathbf{y}(t_0)$, we identify its nearest neighbor $\mathbf{y}(t'_0)$ (excluding temporally adjacent points) and track the divergence:

$$d_j(i) = \|\mathbf{y}(t_j+i) - \mathbf{y}(t'_j+i)\|$$

The largest Lyapunov exponent is estimated from the slope of the average logarithmic divergence:

$$\langle \ln d_j(i) \rangle \approx \lambda_{\max} \cdot i \cdot \Delta t + c$$

where Δt is the sampling interval and c is a constant. We additionally implement the Wolf algorithm²⁹ as a complementary validation method. The Rosenstein method is adopted as the primary estimator throughout this study owing to its ensemble-averaged design, which provides greater statistical robustness for non-stationary series of moderate length; the Wolf algorithm serves as a complementary validation tool with a shorter effective integration window (10 steps vs. 50 steps), making it more sensitive to transient local expansion events.

Phase Space Reconstruction and Poincaré Sections

Phase portraits visualize system trajectories in reconstructed phase space, revealing attractor geometry. For 3-dimensional visualization, we use embedding parameters ($m=3, \tau$) where τ is determined by the first minimum of the mutual information function³¹.

Poincaré sections provide dimensional reduction of phase space dynamics by recording successive intersections of trajectories with a hyperplane $\Sigma = \{\mathbf{x} : h(\mathbf{x}) = c\}$. For our 3D reconstructions, we define $\Sigma : z_1 = \bar{z}_1$ (the mean value) and collect crossing points satisfying $\dot{z}_1 > 0$. The resulting 2D section reveals the underlying structure: scattered points suggest chaos, while organized curves indicate quasi-periodicity².

Return maps plot successive Poincaré section intersections (z_n vs. z_{n+1}), providing additional dynamical characterization. A functional relationship suggests deterministic dynamics, while scattered distributions indicate stochastic or high-dimensional chaotic behavior.

Table 1. Descriptive statistics of daily Tai Chi search index (2014–2024). Note: CV = coefficient of variation; kurtosis values are excess kurtosis.

Province	Channel	N	Mean	SD	Min	Median	Max	Skew.	Kurt.	CV(%)
Beijing	PC	4,018	122.6	48.8	0	112	650	1.69	10.98	39.8
	Mobile	4,018	218.5	84.4	80	199	933	1.99	9.37	38.6
Guangdong	PC	4,018	118.5	41.8	0	110	627	1.59	9.51	35.3
	Mobile	4,018	382.0	168.5	122	345	1,392	0.80	0.63	44.1
Henan	PC	4,018	100.8	36.7	0	89	346	1.25	1.67	36.4
	Mobile	4,018	312.4	113.4	110	299	1,049	1.06	2.68	36.3
Shanghai	PC	4,018	98.8	30.1	0	93	400	1.48	7.15	30.5
	Mobile	4,018	165.7	60.3	66	158	873	2.91	24.41	36.4

Datasets and Processing

This study utilizes daily search index data for the keyword “Tai Chi” (in Chinese) collected from four Chinese provinces over an 11-year period from January 1, 2014 to December 31, 2024. The four provinces—Beijing, Shanghai, Guangdong, and Henan—were selected to represent distinct regional characteristics: Beijing as the political and cultural capital, Shanghai as the leading economic center, Guangdong as the most populous province with a robust digital economy, and Henan as the birthplace of Tai Chi where Chenjiagou Village has served as the spiritual home of this martial art tradition since the mid-17th century. The dataset comprises two channels for each province: PC-based and mobile-based search indices, reflecting user engagement through different digital platforms. Each time series contains 4,018 daily observations, yielding a total of 32,144 data points (4 provinces $\times$ 2 channels $\times$ 4,018 days) for analysis.

Prior to analysis, the raw time series underwent preprocessing to mitigate noise while preserving dynamical features. A 7-day centered moving average filter was applied to smooth high-frequency fluctuations attributable to weekly periodicity and measurement noise. Subsequently, each smoothed series was standardized using z-score normalization: $x'_t = (x_t - \mu)/\sigma$, where μ and σ denote the series mean and standard deviation, respectively. This standardization ensures comparability across provinces and channels with different absolute magnitudes while preserving the relative temporal dynamics essential for HVG construction and dynamical analysis.

Results

Statistical Profiling

Table 1 presents the descriptive statistics for the daily Tai Chi search index across four provinces and two channels, based on a dataset of 32,144 observations spanning from January 1, 2014, to December 31, 2024. Guangdong exhibits the highest mobile search volume, with a mean of 382.0 and a standard deviation of 168.5, approximately 2.3 times that of Shanghai, which has a mean of 165.7 and a standard deviation of 60.3. This disparity reflects differences in population size and the digital economy between the two regions. Mobile search indices consistently exceed those of PC-based searches across all provinces, with Mobile/PC ratios ranging from 1.68 in Shanghai to 3.22 in Guangdong, followed closely by Henan at 3.10. These findings underscore a shift toward mobile-driven search behavior in the studied regions.

Table 2. Annual mean total search index and temporal change. Note: $\Delta(\%)$ = percentage change from peak year to 2024.

Year	Beijing	Guangdong	Henan	Shanghai
2014	415	599	462	293
2015	466	758	573	359
2016	493	704	519	338
2017	452	712	548	348
2018	388	571	456	295
2019	318	478	389	262
2020	292	417	375	228
2021	256	358	338	216
2022	243	370	367	221
2023	226	283	273	182
2024	202	256	244	168
$\Delta(\%)$	−59.0	−66.2	−57.4	−53.2

The time series data for all provinces display positive skewness (ranging from 0.80 to 2.91) and excess kurtosis (ranging from 0.63 to 24.41), indicating right-skewed distributions with heavy tails. These characteristics are typical of attention-driven phenomena, where sporadic events lead to transient surges in public interest, resulting in highly volatile search patterns. The annual trend analysis, as shown in Table 2, reveals a consistent decline in total search volume from the peak years (2015–2016) to 2024. The reductions range from 53.2% in Shanghai to 66.2% in Guangdong, reflecting a broader downward trend in search activity. Figure 1 visualizes the raw time series, annual trends, and the evolution of the Mobile/PC ratio, highlighting both the overall decline in search volume and the accelerating shift toward mobile-based information seeking over the study period.

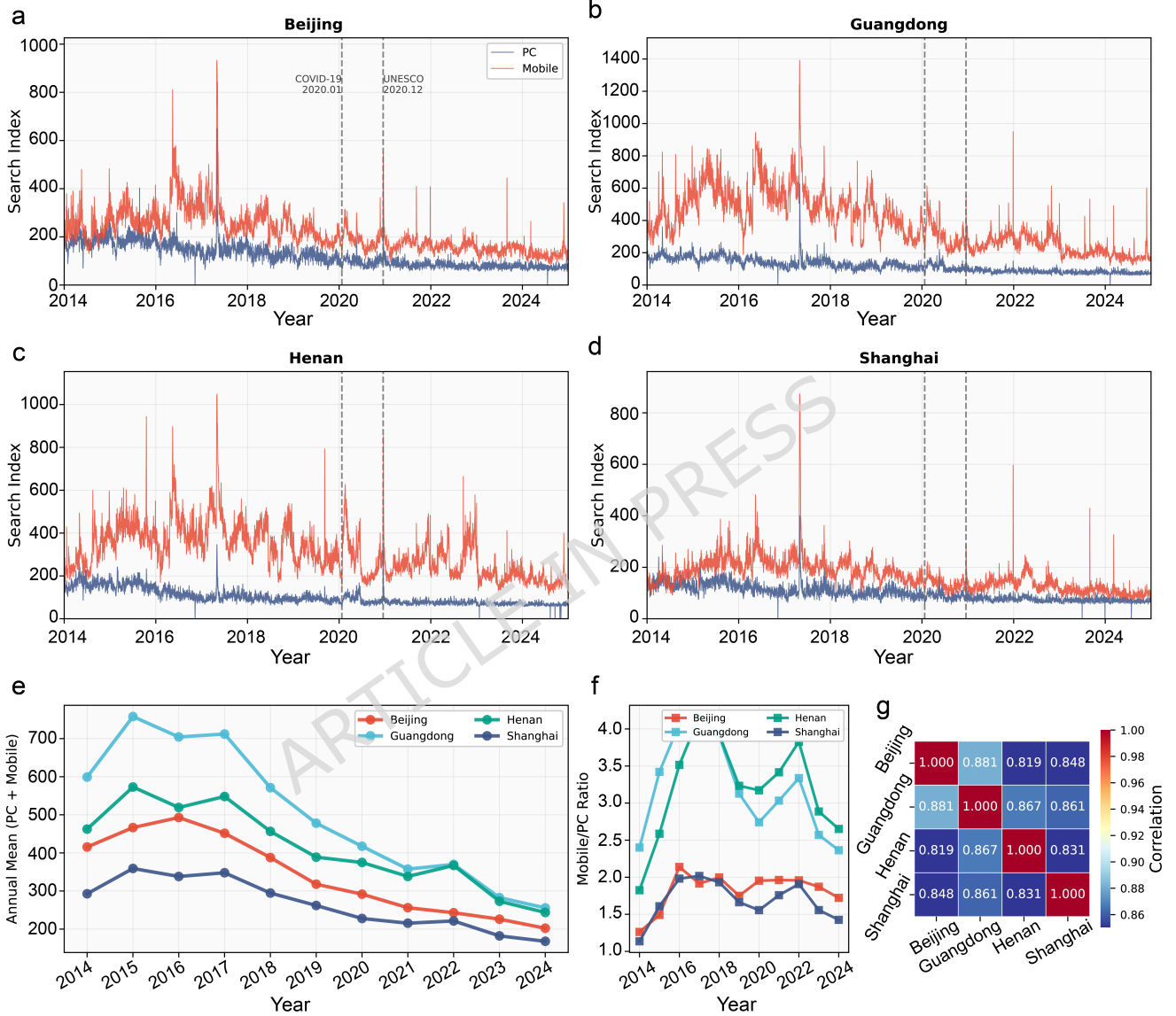


Figure 1. Temporal patterns of Tai Chi search index. (a)–(d) Raw time series for Beijing, Guangdong, Henan, and Shanghai, respectively (PC: blue; Mobile: red; key events annotated: UNESCO inscription 2020.12, COVID-19 2020.01). (e) Annual mean trends by province. (f) Evolution of Mobile/PC ratio (2014–2024). (g) Cross-province correlation heatmap.

HVG-Lyapunov Chaos Analysis

The Horizontal Visibility Graph methodology, when applied to the preprocessed Baidu Index time series, uncovers substantial heterogeneity in the underlying dynamical regimes across provinces and digital platforms. Table 3 summarizes the degree distribution characteristics derived from the constructed HVGs. The exponential decay parameter λ , which serves as a discriminating statistic for dynamical classification, spans a considerable range from 0.257 (Shanghai Mobile) to 0.437 (Shanghai PC), with coefficient of determination values ($R^2 = 0.431$ – 0.558) on the semi-logarithmic scale and regression

Table 3. Chaos characterization: HVG and Lyapunov exponent analysis. Note: Reference HVG threshold $\ln(3/2) \approx 0.4055$; Lyapunov units = bits/day; embedding: $m = 5$, $\tau = 7$.

Province	Channel	HVG λ	R^2	p -value	$\lambda_{\max}$ (Rosen)	$\lambda_{\max}$ (Wolf)	Chaos
Beijing	PC	0.291	0.431	0.0108	0.027	0.171	Strong
	Mobile	0.316	0.531	0.0002	0.033	0.179	Strong
Guangdong	PC	0.376	0.449	0.0087	0.026	0.020	Strong
	Mobile	0.376	0.539	0.0005	0.034	0.174	Strong
Henan	PC	0.415	0.538	0.0019	0.026	0.0164	Weak
	Mobile	0.434	0.558	0.0009	0.036	0.217	Weak
Shanghai	PC	0.437	0.438	0.0062	0.025	0.195	Weak
	Mobile	0.257	0.536	0.0013	0.032	0.178	Strong

slope p -values of $p < 0.05$ in all eight cases (seven of eight satisfy $p < 0.01$, with the exception of Beijing PC at $p = 0.0108$), confirming statistically significant exponential decay throughout. The modest R^2 values are consistent with expectations for finite observational time series exhibiting non-stationary trends, as log-transformation compresses the dynamic range of $P(k)$ and finite-size effects introduce scatter around the theoretical form; crucially, the chaos classification is triangulated across three independent metrics—HVG λ , Rosenstein $\lambda_{\max}$, and Wolf $\lambda_{\max}$ —and does not rest on the regression fit quality alone.

Applying the established HVG chaos criterion—where $\lambda < \ln(3/2) \approx 0.4055$ demarcates chaotic or correlated stochastic processes from more regular periodic dynamics—reveals a clear partition among the eight time series. Five series (Beijing PC/Mobile, Guangdong PC/Mobile, and Shanghai Mobile) fall below this critical threshold, providing evidence of deterministic chaos, while the remaining three (Henan PC/Mobile and Shanghai PC) exceed the threshold, suggesting quasi-periodic behavior or weaker chaotic signatures. This finding establishes that public attention toward Tai Chi does not follow a universal dynamical pattern but instead exhibits pronounced regional specificity, reflecting the diverse socio-cultural contexts in which this traditional practice is embedded.

Complementary validation through Lyapunov exponent estimation strengthens the evidence for low-dimensional chaos in these attention dynamics (Table 3). All eight series yield strictly positive $\lambda_{\max}$ values under both estimation methods, confirming the presence of sensitive dependence on initial conditions—the hallmark of chaotic systems. Specifically, the Rosenstein algorithm yields values ranging from 0.025 to 0.036 bits/day, while the Wolf algorithm yields a broader range of 0.016 to 0.217 bits/day. Despite the quantitative discrepancy between the two methods, the consistent positivity of $\lambda_{\max}$ across all series and both algorithms provides robust cross-method validation of chaotic dynamics. These magnitudes correspond to predictability horizons of approximately 30–70 days, beyond which forecast skill would deteriorate substantially due to exponential error growth. The two estimators yield quantitatively distinct but qualitatively concordant results: the Rosenstein method produces values of 0.025–0.036 bits/day, reflecting the global mean trajectory separation rate of the reconstructed attractor, while the Wolf method yields a broader range of 0.016–0.217 bits/day, capturing transient local expansion rates that are disproportionately amplified by the spike-driven attention surges characteristic of social media data. Notably, the magnitude of inter-method divergence itself varies across provinces—being particularly pronounced in Shanghai and Henan—suggesting that attractors with more heterogeneous local geometry exhibit greater discrepancy between global and local expansion rates, a pattern consistent with the attractor volume differences observed in Figure 4e. The consistent positivity of $\lambda_{\max}$ across all eight series under both algorithms provides robust cross-method confirmation of chaotic dynamics. The observed inverse correlation between HVG λ and Rosenstein $\lambda_{\max}$ ($r = -0.72$) provides further theoretical validation: lower HVG λ values indicate degree distributions closer to chaotic reference systems, while higher $\lambda_{\max}$ directly quantifies the rate of trajectory divergence. This negative association is expected: lower HVG λ values indicate degree distributions closer to chaotic reference systems, whereas higher Lyapunov exponents directly quantify the rate of trajectory divergence. That both independent metrics converge on consistent chaos classifications across provinces reinforces confidence in the underlying dynamical interpretation.

The integrated visualization in Figure 2 synthesizes these chaos diagnostics while revealing structural patterns not immediately apparent from tabulated metrics alone. The empirical HVG degree distributions conform well to theoretical exponential decay, though subtle tail deviations suggest finite-size effects inherent to observational time series of limited duration (Figure 2a). Lyapunov convergence trajectories display the requisite linear growth region that validates the embedding parameter selection and confirms adequate attractor reconstruction (Figure 2b). More revealing is the joint distribution of HVG λ and Lyapunov $\lambda_{\max}$ across provinces, which exposes distinct regional clustering within the chaos parameter space. Beijing and Shanghai anchor opposite extremes of this dynamical continuum, while Guangdong and Henan occupy intermediate positions—a configuration suggesting that regional information ecosystems impose characteristic signatures on attention dynamics (Figure 2c). The platform comparison yields perhaps the most unexpected finding: whereas Beijing and Henan exhibit stronger chaos in PC channels relative to Mobile ($\Delta\lambda = 0.025$ and 0.019 respectively), Guangdong presents a distinct pattern in which PC and Mobile channels yield identical λ values (both 0.376), indicating dynamical convergence between platforms that likely reflects

the homogenizing effect of Guangdong's highly developed and demographically diverse digital ecosystem on cross-platform attention behavior. Shanghai, by contrast, displays a pronounced reversal with Mobile showing substantially stronger chaotic signatures ($\lambda = 0.257$) than PC ($\lambda = 0.437$). (Figure 2d). This anomalous platform divergence, unique to Shanghai among the provinces examined, implies fundamentally different user engagement regimes that likely reflect the city's distinctive demographic composition and digital consumption culture.

The regional hierarchy revealed by the chaos metrics offers insights into how Tai Chi attention evolves across different cultural and geographic contexts. The provincial dynamical structure can be characterized directly from the channel-level λ values and their positions relative to the $\ln(3/2)$ threshold. Beijing exhibits the strongest chaotic signatures across both channels (PC: $\lambda = 0.291$, Mobile: $\lambda = 0.316$), with both values ranking lowest among all eight series and lying well below the threshold, indicating that strong chaos is a robust feature of Tai Chi attention dynamics in the capital regardless of access platform. Guangdong displays remarkable dynamical convergence between platforms ($\lambda_{PC} = \lambda_{Mobile} = 0.376$), with both channels falling below the threshold at identical values—a pattern that distinguishes Guangdong from all other provinces and likely reflects the maturation of its digital ecosystem to a point where the behavioral distinction between desktop and mobile search has effectively dissolved. Henan occupies the opposite extreme: it is the only province where both channels exceed the $\ln(3/2)$ threshold (PC: $\lambda = 0.415$, Mobile: $\lambda = 0.434$), with the two values clustering closely above the boundary, indicating that quasi-periodic dynamics characterize Tai Chi attention in its birthplace across both search platforms and therefore constitute a genuinely provincial rather than platform-specific feature. Shanghai presents the most structurally complex pattern among the four provinces: the PC channel falls above the threshold ($\lambda = 0.437$, quasi-periodic) while the Mobile channel lies well below it ($\lambda = 0.257$, the lowest value among all eight series), a cross-platform divergence in dynamical regime that is unique among the provinces examined. The consistency between PC and Mobile classifications in Beijing, Guangdong, and Henan reinforces confidence that the observed dynamical regimes reflect genuine provincial characteristics rather than platform-specific measurement artifacts, while Shanghai's divergence itself constitutes a substantive finding about the heterogeneous information ecosystems operating within a single province.

Henan, the birthplace of Tai Chi, exhibits quasi-periodic attention dynamics across both search platforms, with attention closely following predictable rhythms possibly aligned with traditional practice schedules and cultural events. In contrast, Beijing shows the strongest chaos across both channels, reflecting the pervasive influence of political, media, and institutional factors in the capital that continuously reshape public attention trajectories. Guangdong and Shanghai, both economically advanced coastal regions, represent intermediate states where traditional cultural patterns interface with the unpredictable dynamics of modern digital media, though in distinct ways: Guangdong's platform-invariant moderate chaos suggests a regionally homogeneous digital engagement culture, whereas Shanghai's significant platform divergence—strong chaos in Mobile and quasi-periodicity in PC—suggests generational or occupational differences in engagement, with younger mobile users more prone to viral social media effects.

A conceptually important distinction emerges from the juxtaposition of Lyapunov exponents and HVG λ values across provinces. The Rosenstein λ_{max} estimates are broadly similar across all regions (0.025–0.036 bits/day), implying comparable short-term predictability horizons of approximately 28–40 days regardless of province. What genuinely differentiates regional dynamics is therefore not predictability over short windows but the *geometric structure* of the underlying attractor: Henan's HVG λ approaching $\ln(3/2)$ and its lower correlation dimension indicate that attention trajectories are organized around quasi-periodic orbits shaped by culturally embedded seasonal and ceremonial rhythms, whereas Beijing's space-filling attractor geometry reflects a system where no single dominant rhythm prevails and perturbations from diverse institutional sources continuously reshape the trajectory. This distinction—between systems that are equally unpredictable over short horizons yet fundamentally different in their long-term phase space organization—underscores the value of combining multiple chaos metrics rather than relying on Lyapunov exponents alone. This perspective resonates with broader work on resilience and early-warning signals in human behavioral systems, where the geometry of attractors has been shown to encode meaningful information about system stability and recovery capacity²¹.

Latent Dynamics Discovery

The autoencoder compresses the 12-dimensional HVG feature vectors into a 3-dimensional latent space, effectively preserving the underlying dynamical structure. Table 4 reports the reconstruction performance, demonstrating that the standard autoencoder achieves normalized mean squared errors ranging from 20.0% to 23.1% across provinces, with correlation coefficients between the original and reconstructed sequences ranging from 0.896 to 0.918. In contrast, the VAE variant produces significantly higher errors (32.4%–32.7%) and lower correlations (0.821–0.823), although it generates smoother latent manifolds. Given the superior reconstruction performance of the standard autoencoder, subsequent SINDy analysis primarily utilizes the latent representations derived from the autoencoder.

The latent dynamics of Tai Chi attention across provinces are captured through SINDy (Figure 3), revealing distinct regional patterns in the 3D latent space (Figure 3a). Henan and Beijing cluster closely, reflecting shared cultural influences that likely

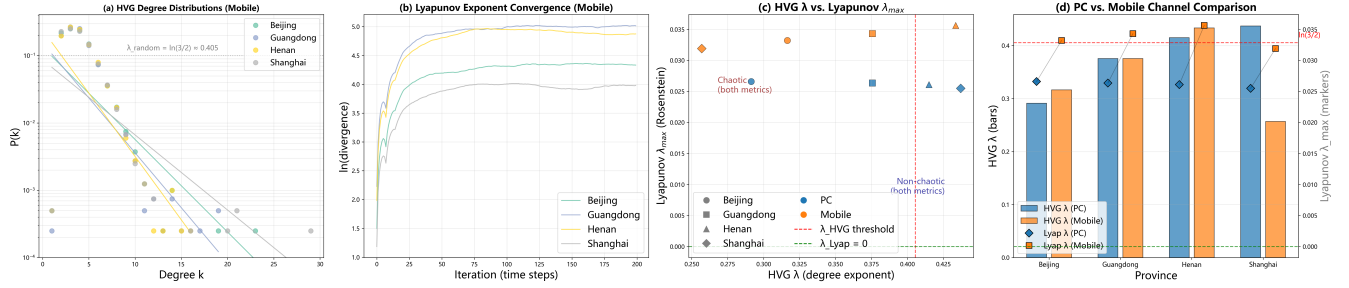


Figure 2. Integrated chaos characterization. (a) HVG degree distributions $P(k)$ with exponential fits for four provinces (Mobile channel). (b) Lyapunov exponent convergence curves showing average logarithmic divergence vs. Iteration (Mobile channel). (c) Scatter plot of HVG λ vs. Lyapunov $\lambda_{\max}$ with provincial labels (shapes: province; colors: channel). (d) Paired comparison of chaos metrics between PC and Mobile channels.

Table 4. Autoencoder reconstruction performance. Note: Error = normalized MSE; Correlation = Pearson r between original and reconstructed feature sequences.

Province	AE Error (%)	VAE Error (%)	Correlation (AE)	Correlation (VAE)
Beijing	20.6	32.6	0.910	0.821
Guangdong	23.1	32.7	0.896	0.822
Henan	22.2	32.7	0.902	0.821
Shanghai	20.0	32.4	0.918	0.823

synchronize attention dynamics with traditional rhythms and local events, while Guangdong's position in a high- z region suggests a unique dynamical structure, potentially shaped by its distinct socio-economic context. Shanghai forms the most compact cluster, indicating a more homogeneous dynamic, possibly influenced by rapid urbanization and media-driven attention. Temporal latent trajectories (Figure 3b) exhibit non-periodic, bounded evolution typical of strange attractors, highlighting the chaotic yet constrained nature of attention dynamics, driven by factors such as media exposure and cultural events. Prediction accuracy for Beijing's latent trajectories (Figure 3c) shows good alignment between actual and predicted values, further validating the robustness of the latent representation. The SINDy coefficient magnitude heatmap (Figure 3d) reveals notable regional variations, with Beijing exhibiting the highest coefficients in constant and trigonometric terms, suggesting that cyclical and cultural factors play a more prominent role in its attention dynamics. Conversely, Guangdong and Henan show comparatively lower magnitudes, indicating less influence from periodic or media-driven dynamics.

The discrete SINDy algorithm identifies sparse governing equations from the latent trajectories. As a representative example, the dynamics identified for Beijing are as follows:

$$\begin{bmatrix} z_1(t+1) \\ z_2(t+1) \\ z_3(t+1) \end{bmatrix} = C + A \underbrace{\mathbf{z}}_{\text{linear}} + \underbrace{\begin{bmatrix} \mathbf{z}^\top Q_1 \mathbf{z} \\ \mathbf{z}^\top Q_2 \mathbf{z} \\ \mathbf{z}^\top Q_3 \mathbf{z} \end{bmatrix}}_{\text{quadratic}} + D \underbrace{\phi}_{\text{trigonometric}}, \text{ where } \mathbf{z} = [z_1 \ z_2 \ z_3]^\top, \phi = [\sin z_1 \ \cos z_1 \ \sin z_2 \ \cos z_2 \ \sin z_3 \ \cos z_3]^\top$$

and the matrix values are as follows:

$$C = \begin{bmatrix} 1.7184 \\ 1.0334 \\ -0.9239 \end{bmatrix}, \quad A = \begin{bmatrix} 0.2143 & -0.2071 & -0.3655 \\ 0.5407 & -0.4117 & 0 \\ -0.0810 & 0.7964 & -0.4133 \end{bmatrix},$$

$$D = \begin{bmatrix} -0.1041 & -0.5770 & 0.2426 & -0.5909 & 0.5289 & -0.5028 \\ -0.4644 & 0 & 0.6341 & -1.0205 & 0 & 0 \\ 0 & 0 & -0.6598 & 2.0008 & 0.5834 & -1.0934 \end{bmatrix},$$

$$Q_1 = \begin{bmatrix} -0.1827 & -0.0275 & 0 \\ -0.0275 & -0.1588 & 0 \\ 0 & 0 & -0.2890 \end{bmatrix}, \quad Q_2 = \begin{bmatrix} -0.0857 & -0.0743 & 0.0254 \\ -0.0743 & -0.3604 & -0.0342 \\ 0.0254 & -0.0342 & 0 \end{bmatrix}, \quad Q_3 = \begin{bmatrix} 0 & 0.0875 & 0 \\ 0.0875 & 0.5778 & 0.0444 \\ 0 & 0.0444 & -0.4634 \end{bmatrix}.$$

Table 5 compares the SINDy coefficient structures across provinces. The identified dynamics include linear terms with mean coefficients ranging from 0.222 (Henan) to 0.398 (Guangdong), along with quadratic terms (0.092–0.150) and trigonometric terms (0.287–0.500) that introduce the nonlinearity responsible for chaotic behavior. Notably, Beijing’s equations exhibit the strongest trigonometric components (mean = 0.500), while Henan shows the weakest (mean = 0.287), which contrasts with Henan’s more periodic HVG characteristics. The models retain between 30 and 38 active terms out of a total of 48 library terms, reflecting a moderate level of sparsity (approximately 10–13 active terms per latent dimension). The one-step-ahead prediction yields a normalized root mean square error (NRMSE) ranging from 11.7% to 12.8%, demonstrating reasonable predictive capability given the complexity of the attention dynamics.

Table 5. SINDy coefficient structure comparison across provinces. Note: Mean values represent average absolute coefficient magnitudes; Active Terms = nonzero coefficients / total library size (19 candidates $\times$ 3 dimensions).

Province	Linear (mean)	Quadratic (mean)	Trigonometric (mean)	Active Terms	NRMSE (%)
Beijing	0.337	0.150	0.500	37/48	12.7
Guangdong	0.398	0.111	0.354	38/48	11.9
Henan	0.222	0.115	0.287	38/48	12.8
Shanghai	0.240	0.092	0.349	30/48	11.7

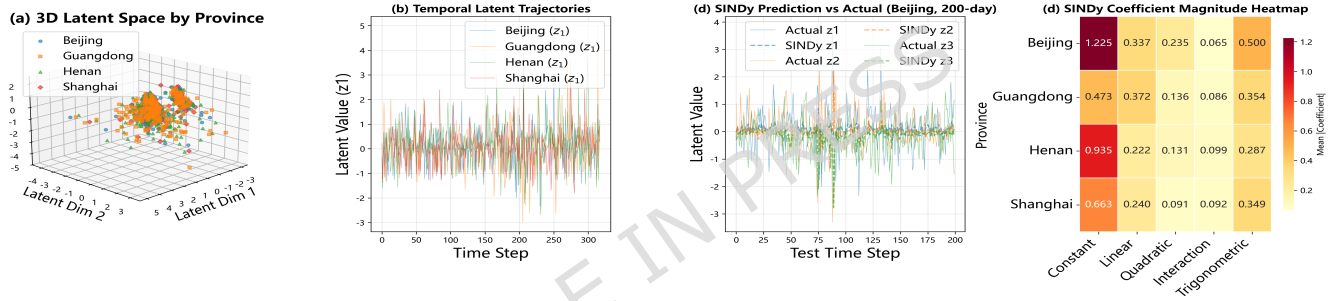


Figure 3. Latent space learning and SINDy identification. (a) 3D latent space visualization colored by province. (b) Temporal latent trajectories for each province. (c) SINDy prediction vs. actual latent values for Beijing (200-day test period). (d) Coefficient magnitude heatmap across provinces and term categories.

Attractor Geometry Characterization

Phase space reconstruction using delay embedding ($m = 3$, $\tau = 7$ days) reveals distinct attractor geometries that capture the underlying dynamics of Tai Chi attention across provinces. The 3D phase portraits (Figure 4a,b) show bounded, non-periodic trajectories, characteristic of strange attractors, with notable regional differences. Henan’s attractor displays quasi-periodic bands, suggesting weaker chaotic dynamics likely influenced by more stable and predictable cultural rhythms, while Beijing exhibits a more uniform space-filling structure, indicative of stronger chaotic mixing, consistent with its lowest HVG λ values. Poincaré sections (Figure 4c) further validate the presence of low-dimensional deterministic chaos, with structured point distributions confirming the deterministic nature of the dynamics. Henan’s return maps (Figure 4d) show tight clustering, pointing to periodic behavior, while Beijing’s greater dispersion reflects the more chaotic nature of its attention dynamics.

The correlation dimension (Figure 4e) estimates align with the regional differences observed in the HVG analysis, with Henan showing the lowest correlation dimension, indicating weaker chaos, and Beijing exhibiting the highest dimension, reflecting its stronger chaotic characteristics. Additionally, attractor volumes, as measured by the convex hull, correlate positively with chaos strength, suggesting that regions with greater chaotic dynamics, such as Beijing, have larger attractor volumes, indicative of more unpredictable attention behaviors. The SINDy coefficient magnitude heatmap (Figure 3e) provides further insight into the regional dynamics, showing that Beijing’s attention dynamics are most influenced by nonlinear, trigonometric components, highlighting the role of periodic and media-driven factors in its chaotic behavior. In contrast, Guangdong and Henan exhibit lower SINDy coefficient magnitudes, consistent with their more structured phase space geometry and quasi-periodic dynamical tendencies as characterized by the HVG analysis, thus reinforcing the regional heterogeneity in the nature of Tai Chi attention.

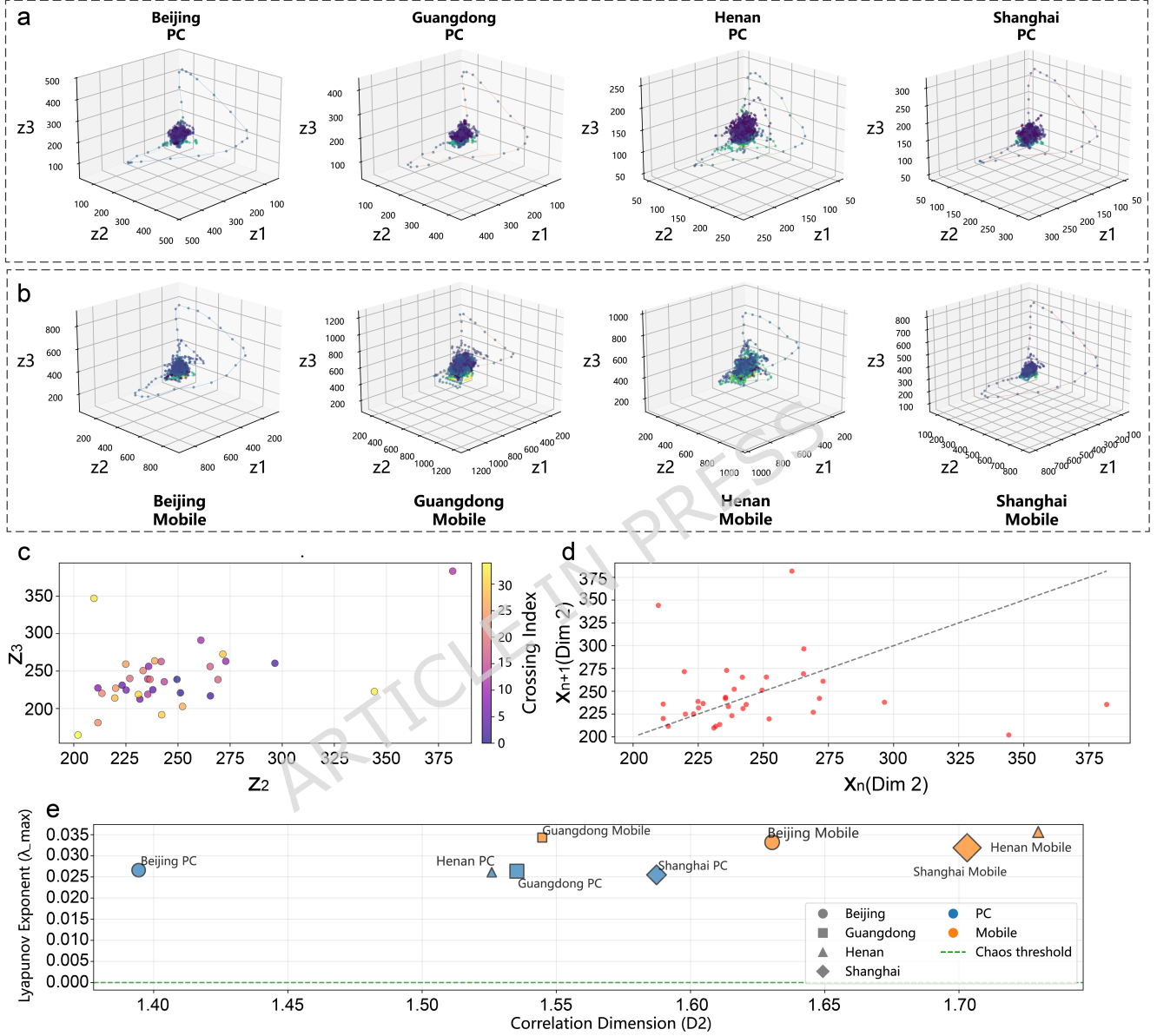


Figure 4. Phase space geometry and attractor structure. (a,b) 3D phase portraits for eight time series (PC and Mobile). (c) Poincaré sections at $z_1 = \text{mean}(z_1)$ (Beijing Mobile). (d) Return maps (z_n vs. z_{n+1}) with local regression fits (Beijing Mobile). (e) Attractor metric comparison: correlation dimension vs. Lyapunov exponent with convex hull volume indicated by marker size.

Comparative Analysis of Dynamical Modeling Approaches (Mobile Channel, 50-day Test Period)

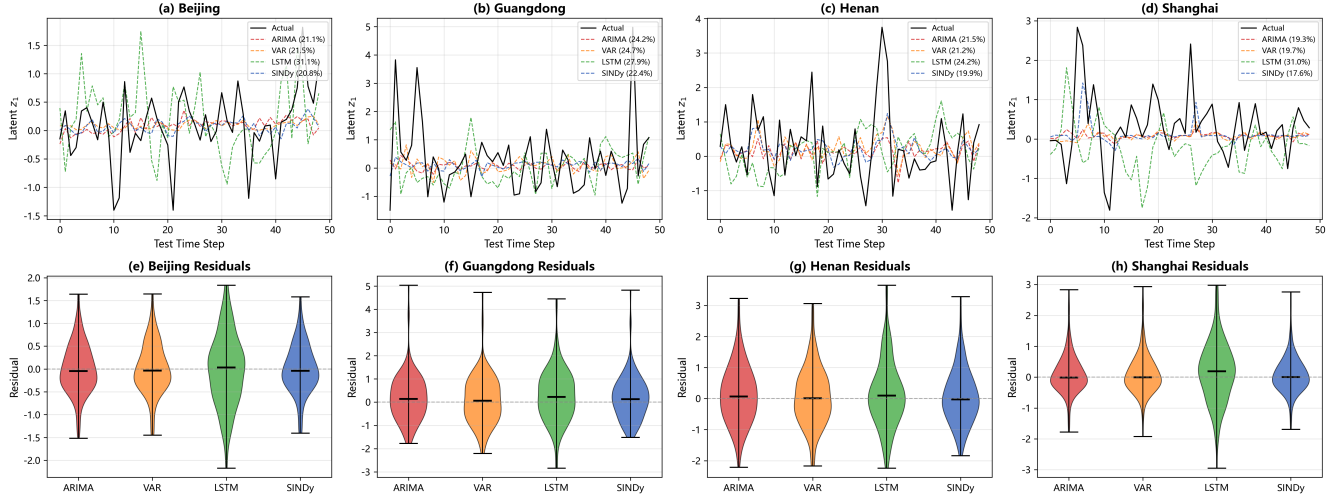


Figure 5. Comparative analysis of four dynamical modeling approaches applied to Mobile channel latent trajectories across four provinces (50-day test period). (a)–(d) One-step-ahead prediction curves for Beijing, Guangdong, Henan, and Shanghai respectively; black lines denote actual latent values z_1 , colored dashed lines denote ARIMA (red), VAR (orange), LSTM (green), and SINDy (blue), with NRMSE values reported in the legend. (e)–(h) Residual violin plots for the corresponding provinces, showing the distribution of prediction errors across all three latent dimensions; the dashed horizontal line indicates zero residual.

Comparative Analysis of Dynamical Modeling Approaches

To contextualize the performance of the proposed framework, we compare SINDy against three alternative approaches applied within the same autoencoder latent space: univariate ARIMA, multivariate Vector Autoregression (VAR), and Long Short-Term Memory networks (LSTM). All methods receive identical 3-dimensional latent trajectories as input and are evaluated on a common 50-day test period using Mobile channel data from all four provinces, ensuring that performance differences reflect modeling capacity rather than differences in input representation. Results are presented in Figure 5.

SINDy achieves the lowest one-step-ahead NRMSE across all provinces (Beijing: 20.8%, Guangdong: 22.4%, Henan: 19.9%, Shanghai: 17.6%), outperforming ARIMA (21.1%–24.2%), VAR (21.2%–24.7%), and LSTM (24.2%–31.1%). Notably, LSTM yields the highest prediction errors across all provinces despite its greater model complexity, consistent with the known susceptibility of recurrent networks to overfitting under non-stationary conditions with moderate data volumes. The close performance of ARIMA and VAR confirms that linear multi-dimensional extensions provide limited additional benefit over univariate linear baselines for these chaotic series.

The residual violin plots (Figure 5(e)–(h)) further reveal that SINDy produces the most concentrated residual distributions across all provinces, with narrower interquartile ranges and less extreme outliers compared to the three alternative methods, reflecting more consistent error behavior across the test period. While ARIMA and VAR achieve competitive accuracy on smoother series such as Shanghai, they are structurally incapable of capturing the nonlinear cross-dimensional interactions that SINDy identifies through sparse equation discovery. Beyond predictive performance, SINDy uniquely provides interpretable governing equations whose coefficient structure encodes province-specific dynamical properties, enables chaos detection through Lyapunov analysis, and characterizes attractor geometry — capabilities that neither linear methods nor black-box sequence models can replicate.

Discussion

This study introduces an integrated HVG-SINDy Autoencoder framework that bridges network-based complexity analysis with data-driven dynamical modeling, applied here to characterize the temporal evolution of public attention toward Tai Chi across four Chinese provinces over an 11-year period. The convergence of multiple analytical approaches—visibility graph topology, deep representation learning, and sparse equation discovery—yields a coherent picture of cultural attention as a low-dimensional chaotic system exhibiting pronounced regional specificity.

That the SINDy algorithm successfully identifies sparse governing equations from these latent trajectories validates the premise that cultural attention, despite its apparent complexity, obeys discoverable dynamical laws—a finding that positions

such phenomena within the broader class of systems amenable to equation-free modeling approaches developed in physics and engineering contexts. The dynamical conclusions of this study derive their credibility from the convergence of three analytically independent characterizations: the HVG λ criterion, grounded in complex network theory and sensitive to the topological structure of temporal correlations^{8,22}; the Rosenstein and Wolf $\lambda_{\max}$ estimates, grounded in delay embedding theory³² and ergodic theory³³ respectively, and sensitive to the geometric divergence of phase space trajectories; and the SINDy-identified governing equations, whose regional variation in coefficient structure mirrors the dynamical hierarchy revealed by the network and phase space analyses. The parallel training of standard and variational autoencoders further provides an internal cross-check on the latent representation: that both architectures recover provincially coherent clustering without supervisory signal suggests that the learned coordinates reflect genuine dynamical structure rather than architectural artefacts. The mutual consistency of these independent lines of evidence — each rooted in a distinct theoretical tradition yet converging on the same regional hierarchy of chaos strength — constitutes the primary warrant for the dynamical interpretations advanced in this study.

The comparative analysis against ARIMA, VAR, and LSTM further substantiates the framework's practical value. That SINDy achieves the lowest prediction error across all four provinces despite operating under identical input conditions confirms that sparse nonlinear equation discovery captures genuine dynamical structure rather than merely fitting noise. The counterintuitive underperformance of LSTM relative to simpler linear baselines is itself a dynamically meaningful finding: it reflects the sensitivity of recurrent black-box models to the non-stationarity and moderate effective sample sizes characteristic of decade-long cultural attention series, and serves as a cautionary note for uncritical application of deep sequence models to social time series without prior dynamical characterization.

The central finding that Tai Chi attention dynamics partition into chaotic and quasi-periodic regimes across provinces challenges prevailing assumptions in cultural analytics. Conventional approaches typically treat public attention as either purely stochastic or driven by identifiable external events, yet our results suggest an intermediate regime: deterministic chaos with predictability horizons of 30–70 days. This timescale carries practical implications for cultural heritage management, as it defines the window within which promotional interventions or policy initiatives might achieve predictable outcomes before intrinsic dynamical instabilities dominate. The regional heterogeneity we observe—with Henan displaying quasi-periodic behavior while Beijing exhibits strong chaos—likely reflects fundamental differences in how traditional culture interfaces with modern information ecosystems. In its birthplace, Tai Chi attention appears anchored to stable cultural institutions and practice communities that impose regular temporal structure; in the national capital, attention dynamics are shaped by a more volatile confluence of policy signals, media amplification, and competing cultural narratives.

These regional differences are also broadly consistent with provincial population size and urbanization profiles. Guangdong, as China's most populous province with a highly developed digital economy, exhibits an intermediate chaos profile that may reflect the superposition of large-scale urban mobile engagement atop a residual base of traditional practice communities. Henan, by contrast, despite its substantial population, maintains a more culturally homogeneous relationship with Tai Chi as its birthplace, which may suppress the disruptive influences that amplify chaos in more cosmopolitan settings. A rigorous demographic analysis of these relationships, however, lies beyond the scope of the present study and represents a valuable direction for future investigation.

The SINDy-identified governing equations merit careful interpretation. The moderate sparsity (30–38 active terms from a 48-term library) indicates that attention dynamics, while reducible to tractable mathematical form, resist extreme simplification. Particularly noteworthy is the inverse relationship between HVG-derived periodicity and trigonometric term strength in the discovered equations: Henan's quasi-periodic attention patterns arise not from dominant harmonic components but from complex nonlinear interactions among lower-order terms. This counterintuitive finding suggests that cultural rootedness manifests dynamically through intricate feedback mechanisms rather than simple seasonal or cyclical drivers—a distinction with implications for understanding how traditional practices maintain temporal coherence across generations. The stronger trigonometric signatures in Beijing's equations may instead reflect externally imposed periodicities from institutional calendars, policy cycles, or media scheduling that overlay the underlying chaotic dynamics.

The platform-dependent asymmetry between PC and Mobile channels reveals an unexpected dimension of regional specificity. The pattern across Beijing and Henan—where desktop-based attention exhibits stronger chaos than mobile—aligns with intuitive expectations that deliberate, search-oriented behavior on computers might generate more complex temporal patterns than passive mobile browsing. Guangdong's platform-invariant dynamics ($\lambda_{\text{PC}} = \lambda_{\text{Mobile}} = 0.376$) suggest a third regime in which the platform distinction has effectively dissolved, possibly reflecting the maturation of Guangdong's mobile ecosystem to a point where mobile search behavior is as deliberate and topically focused as desktop search, consistent with Guangdong's position as China's leading province in mobile internet penetration and digital economy development. Shanghai's reversal of this pattern demands explanation. We hypothesize that Shanghai's highly developed mobile ecosystem, younger demographic profile, and integration of social media sharing into daily routines may have inverted the typical device-behavior relationship, with mobile platforms serving as primary conduits for active cultural engagement rather than passive consumption. This interpretation, if validated through user-level data, would carry implications for how digital platforms mediate cultural

transmission in economically advanced urban centers.

This study has limitations that constrain interpretation. Reliance on Baidu Index data, while providing consistent longitudinal coverage, captures only search-based attention and excludes social media engagement, offline participation, and international interest. The identified equations remain phenomenological: they describe rather than explain, and the coefficients lack direct correspondence to measurable social or cultural processes. The 11-year observation window, though substantial for digital data, may inadequately sample slow dynamical modes or miss regime transitions induced by structural changes in China's digital landscape. Furthermore, our four-province sample, while spanning diverse regional contexts, cannot establish whether the observed patterns generalize across China's remaining provinces or to other intangible cultural heritage domains.

From a methodological standpoint, the systematic quantitative discrepancy between Rosenstein and Wolf estimates observed here carries a broader lesson for chaos analysis of social time series: the Wolf method's sensitivity to transient high-divergence events—precisely the spike-like attention surges characteristic of culturally driven phenomena—renders it susceptible to overestimation of $\lambda_{\max}$ in the presence of non-stationarity, while the Rosenstein method's ensemble averaging provides more conservative and interpretable estimates. Future studies applying Lyapunov-based chaos detection to social or cultural attention data are therefore advised to report both estimators and to treat their discrepancy as additional diagnostic information about attractor heterogeneity rather than as mere methodological inconsistency.

The comparative analysis presented here, while demonstrating SINDy's advantages within the autoencoder latent space, evaluates only one-step-ahead prediction accuracy and residual distributions; multi-step forecast horizons and out-of-distribution generalization remain untested. Furthermore, the LSTM baseline was trained on moderate-length latent trajectories without extensive hyperparameter optimization, and its underperformance relative to linear methods should be interpreted with this constraint in mind rather than as a general indictment of recurrent architectures for cultural attention modeling.

Future extensions should pursue mechanistic grounding by incorporating exogenous covariates—policy interventions, media events, demographic shifts—into the SINDy framework, potentially revealing how external perturbations propagate through attention dynamics. Real-time implementation could enable anomaly detection for cultural heritage monitoring, flagging unusual attention patterns that might signal emerging threats or opportunities. Comparative application to other traditional practices—martial arts, traditional medicine, folk customs—would test whether the chaotic-periodic spectrum and regional heterogeneity observed here represent universal features of cultural attention or Tai Chi-specific phenomena. Finally, integration with agent-based modeling could bridge the gap between our phenomenological equations and the micro-level social.

Conclusions

In conclusion, this study demonstrates that the temporal evolution of public attention toward traditional cultural practices like Tai Chi constitutes a complex dynamical system amenable to rigorous mathematical characterization. The HVG-SINDy Autoencoder framework offers a powerful toolkit for uncovering the hidden dynamical structures underlying apparently irregular time series, with implications extending beyond cultural heritage analytics to broader applications in social dynamics, public health surveillance, and complex systems science. The comparative evaluation against ARIMA, VAR, and LSTM baselines further confirms that the SINDy component achieves superior one-step prediction accuracy while simultaneously providing interpretable governing equations and chaos diagnostics that black-box and linear alternatives cannot deliver, establishing the integrated framework as both practically competitive and scientifically more informative than existing approaches.

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Author contributions statement

C.Z. designed this research. Y. K. and P. L. participated in various experimental parts of the project implementation, analyzed the experimental results, and wrote the manuscript. L. T. revised the manuscript. All authors reviewed and approved the final manuscript.

Data and code availability

The datasets and codes of this study are accessible on <https://github.com/wind-whispered/HVG-SINDy-Autoencoder>.

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